James Makhlouf

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Mechatronics Engineering Student

Fourth-year Mechatronics Engineering student with a perfect GPA and Dean's List recognition. Strong interest in control systems, robotics, and brain-computer interfaces (BCI), with hands-on experience in research, embedded systems, and software development. Actively engaged in interdisciplinary projects bridging mechanical, electrical, and computational engineering.

Education

Lebanese American University — Byblos, Lebanon
Bachelor in Mechatronics Engineering — *Aug 2023 – May 2028*

- Dean's Distinguished List
- GPA: 4.0/4.0

Experience

Robotics Intern — Jul 2026 – Aug 2026
inmind.ai — *Mkalles, Lebanon*

- Built SLAM simulation pipelines and multi-node ROS 2 packages in Dockerized environments.
- Developed a simulated delivery robot in C++ with ROS 2 and Nav2, featuring cancellable jobs and behavior-tree mission logic.
- Trained Faster R-CNN and YOLO models in PyTorch for warehouse object detection, applying model pruning and ensembling.

Engineering Intern — Jun 2026 – Aug 2026
Mitsulift — *Dbayeh, Lebanon*

- Drafted 2D AutoCAD engineering drawings, including proposal layouts, shop drawings, and installation diagrams.
- Performed Quality Control (QC) field inspections on installed elevator systems, auditing mechanical integrity and logging snag reports.
- Designed a detailed 3D elevator and counterweight assembly in SOLIDWORKS, configuring mechanical mates to simulate cable motion.
- Constructed a functional 2-floor prototype elevator controlled by an Arduino Mega with IR sensors, limit switches, and motor actuators.

Undergraduate Research — Dec 2025 – Present
Lebanese American University — *Byblos, Lebanon*

- Conducting research in Brain-Computer Interfaces (BCI), focusing on multi-channel EEG signal preprocessing.

Projects

T.R.A.C.E. (Writing Robotic Arm) — *Arduino, MATLAB/Simulink*

- Designed and 3D-printed a 3-DOF robotic arm.
- Implemented a Task-Space Impedance Controller for precise trajectory tracking.
- Simulated on MATLAB using Simscape.

Nine Lives (Game Development Project) — *Flutter (Dart), Flame Game Engine, Firebase*

- Developed a multiplayer and singleplayer pixel-art platformer using Flutter and the Flame engine.
- Implemented Firebase real-time database for real-time multiplayer gameplay.
- Hosted on https://nine-lives-game.web.app/

Inverted Pendulum Control and Swingup — *MATLAB/Simulink*

- Designed and implemented swing-up and stabilization controllers by applying PID, fuzzy logic, LQR, and energy-based control strategies.

Explorer Bot — *Arduino, myRIO*

- Developed an award-winning glove-controlled robotic system featuring a mechanical grabber, camera, and multiple sensors.

Roommate Finder — *Node.js, Express, JavaScript, CSS, HTML*

- Designed and developed a web platform to match users with compatible roommates based on lifestyle, location, and budget.

Technical Skills & Soft Skills

- **Engineering Tools:** MATLAB/Simulink, SOLIDWORKS, AutoCAD, ROS 2, CMake, Docker
- **Programming and Development:** C++, Python, Java, Flutter (Dart), JavaScript, HTML, CSS, PostgreSQL, Firebase
- **Machine Learning:** Deep Learning, Computer Vision, Neural Networks
- **Soft Skills:** Problem Solving, Critical Thinking, Adaptability, Time Management, Teamwork

Certifications & Awards

- Certified SolidWorks Associate (CSWA) by Dassault Systèmes
- LAU Engineering Projects Day Winner (Explorer Bot)
- 1st Place in the Lebanese National Mathematics Olympiad by CRDP

Volunteering

- LAU Robotics Summer Camp 2025: Mentored high school students during hands-on robotics workshops.

Languages

- **English:** C2 | **French:** C1 | **Arabic:** C1